

MANISH DAS

Full Stack Developer | React 18 | Flask | Python | MySQL | TensorFlow

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Location : - Hooghly, West Bengal

GitHub : - github.com/Manish9163

PROFESSIONAL SUMMARY

Full stack developer with two production-shipped projects and a hands-on ML internship. Built FarmFi — an agri-tech PWA (React 18 + Vite + Flask + MySQL + TensorFlow) achieving 94.2% accuracy across 38 crop disease classes, with Razorpay payments, Twilio MFA, JWT-secured REST API design, ACID-compliant inventory locking, and OpenCV Explainable AI (XAI). Experienced in Context API state management, responsive UI with Tailwind CSS and GSAP, and multi-module full-stack architecture. Pursuing BCA at Brainware University (expected 2027); actively seeking full-time roles and open to relocation.

EXPERIENCE

Freshosoft · Machine Learning Intern

1-oct-2025 — 31-dec-2025

- Trained and evaluated CNN image classification models using TensorFlow/Keras and Python — built hands-on understanding of image preprocessing pipelines (resizing, normalization, tensor conversion) and dense network architecture; applied these techniques directly to plant disease detection module, which reached 94.2% accuracy in production.

PROJECTS

FarmFi — AI Agri-Tech Ecosystem · React 18 · Flask · MySQL · TensorFlow · Razorpay

github.com/Manish9163/FarmFi-an_AI_ecosystem

- Engineered a full-stack agri-tech PWA across 50+ React 18 components (Vite + React Router v6 + Flask + MySQL), achieving 45ms auth latency and 120ms marketplace checkout — sustained 50 concurrent users at 15% CPU via Waitress WSGI with 8 parallel workers, ACID-compliant transactions, and SELECT FOR UPDATE row-level locking to prevent inventory double-spending.
- Built a TensorFlow/Keras CNN plant disease classifier achieving 93.2% accuracy · 92.8% precision · 91.5% recall across 38 crop disease classes with 1.2s end-to-end inference latency — implemented OpenCV Explainable AI (XAI) saliency maps to visually annotate which leaf regions triggered each prediction, plus an active learning feedback loop for future model retraining.
- Integrated Razorpay (B2C/B2B payments), Twilio SMS OTP + SMTP email (MFA), OpenWeatherMap API, and JWT-secured REST APIs with bcrypt password hashing and Flask-Limiter DoS prevention — microcredit risk engine and labor marketplace projected to raise farmer profit margins by up to 30% and cut harvest losses by up to 20% via early AI diagnosis.

IoT Traffic Speed Control + Smart Street Lighting · Arduino · IoT · Sensors

Award: Best Project — Brainware University, March 2024

- Built a 10-sensor Arduino Uno IoT system for real-time vehicle speed monitoring at a 60 kmph threshold — solar-powered adaptive street lighting cut energy consumption by ~80–85% versus grid-powered alternatives, while speed violations auto-triggered gate closure and camera-based alerts to the nearest police station.

Indian Wonderer — Full-Stack Travel Booking Platform · React · TypeScript · Angular · Next.js · PHP · MySQL

github.com/Manish9163/Indian-Wonderer-Website

- Architected Indian Wonderer across 6 independent codebases — React + TypeScript (travellers), Angular + Express.js (admin panel), Next.js (guide app), PHP + MySQL (backend), shared-libs, services layer — with 3 user roles, 12+ normalized MySQL tables, real-time GPS tracking for active guide trips, and admin-controlled guide approval workflow across 12 Indian destinations.
- Engineered 12+ booking features: an advanced search engine with 6+ filters (price range, departure time slots, seat class, operator multi-select), visual seat map with real-time availability and per-seat pricing (window/aisle/middle), flexible 3-day date comparison, and a 4-tier loyalty program (Silver → Gold → Platinum → Diamond) earning 1 point per ₹100 with instant redemption.
- Implemented end-to-end booking lifecycle — promo code validation engine with 5 pre-loaded codes and best-discount auto-suggestion, 3-tier travel insurance (Basic/Comprehensive/Premium), partial cancellation with dynamic refund calculation, price alerts with email/SMS notifications, and a verified-purchase review system with 4 rating categories (cleanliness, punctuality, staff, overall).

AI Voice Assistant · Python · SpeechRecognition · OpenAI Whisper

- Built a Python desktop voice assistant using SpeechRecognition and OpenAI Whisper to recognize and execute OS-level commands — enabled hands-free control of browser, notepad, and system navigation, eliminating keyboard dependency for routine desktop tasks.

TECHNICAL SKILLS

Frontend: React 18, TypeScript, JavaScript, Tailwind CSS, GSAP, Lenis, Vite, React Router v6, Context API, Recharts, Angular.js, Next.js, HTML, CSS

Backend: Flask, Python, Node.js, Express.js, PHP, REST API Design, JWT Authentication, bcrypt, Flask-Limiter, Flask-CORS, Waitress WSGI

Database: MySQL, ACID Transactions, Row-Level Locking, Connection Pooling, SQL

AI / Machine Learning: TensorFlow, Keras, OpenCV, NumPy, Pillow, CNN, Explainable AI (XAI), scikit-learn, Matplotlib

Integrations: Razorpay API, Twilio API, OpenWeatherMap API, HTML5 Geolocation API, SMTP

Tools & DevOps: Git, GitHub, PWA, Service Workers, Vite Build Optimization, APScheduler

Other: C Language, MS Office (Word, Excel, PowerPoint)

EDUCATION

Brainware University, Kolkata

July 2023 – Expected 2027

Bachelor of Computer Applications (BCA) · CGPA: 8.3 (5th Semester)

Bora Mandhu Sudan High School

2022

Higher Secondary Certificate — Arts Stream · Score: 79%

LANGUAGES

English (Native) · **Hindi** (Native) · **Bengali** (Professional)